

SILAS GAGNON

CONTACT

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Full stack robotics engineer with with a skill set ranging from mechanical design, to embedded software development. I am a student at Worcester Polytechnic Institute on an endless mission to understand the inner workings of everything.

EXPERIENCE

WPI GOAT FAST RACING

WORCESTER POLYTECHNIC INSTITUTE

AUGUST 2024 - PRESENT

WORCESTER, MA

- Worked on a team of 15 plus engineers to build a fully electric racecar for the Formula Hybrid competition.
- Retrofit 400V accumulator to be completely manufactured in house in under 7 weeks
 - ▶ Created manufacturing and assembly process
 - ▶ Assembled full electro-mechanical system
 - ▶ Created tools to assist with assembly (Spot Welding Jig, Press for copper Bus Bars, Current Testing Jig)
- Designed custom PCB wheel speed sensor using analog quadrature hall effect sensors to measure rotation continuously
- Worked on design of safety systems and procedures for high voltages

USGS STUDENT CONTRACTOR

UNITED STATES GEOLOGICAL SURVEY

MAY 2024 - AUGUST 2024

ANN ARBOR, MI

- Contributed to ROSBag data unpacking library to simplify and streamline programmatic data manipulation.
- Standardized conventions for robot position and orientation.
- Implemented 3D Point Cloud data processing and cleaning scripts to colocate a series of point clouds from GPS data.
- Worked on research vessels in the Great Lakes to gather scientific data with under water autonomous vehicles and sonar scanning technology.

FIRST ROBOTICS TEAM CAPTAIN

FIRST ROBOTICS TEAM 3636

JUNE 2022 - JUNE 2023

PORTLAND, OR

- Led a team of 60 high school students in designing and building a 70 pound robot capable of autonomous navigation and localization in 6 weeks.
- Managed prioritization of sub team resources and coordination among design/CAD, electrical, mechanical, coding, and fabrication teams.
- Practiced Reliability Engineering in an “all or nothing” environment.
- Ranked 11th of 150+ teams, earning a spot in the 2023 FIRST World Championship in Houston, TX.
- Taught STEM and problem solving skills to younger students
- Designed a navigation system that allowed for millions of unique paths to be generated and followed autonomously.

PROJECTS

TELOMETER

- Designed and implemented a telemetry protocol and visualization dashboard
- Capable of doing realtime data synchronization over any serial communication protocol (UART, CAN, UDP, Bluetooth, etc)
- Implemented to send values directly as binary with only 2 bytes of overhead per packet
- Used in robotics courses to enable real time data feedback and control loop tuning.
- Implemented the front end dashboard in zig with live 2d and 3d plots and visualizations
- Packaged with the Nix package manager

WHEEL SPEED SENSORS

- Designed analog quadrature hall effect sensors to measure continuous rotation
- Created custom PCB that communicates analog potential over SPI to a microcontroller
- Implemented sensor processing algorithm and filter to calculate rotation speed.

SKILLS

- Programming Languages: C / C + + , Zig, Python, Java, Kotlin, Nix
- Software Libraries: ImGUI, Eigen, ROS, OpenCV, Numpy
- Manufacturing: 3d printing, CNC Router, CNC Mill, Laser Cutter
- Equipment: Oscilloscope, Multimeter, Logic Analyzer, LCR Meter
- Tools: Basic Woodworking, Electrical, and Assembly Tools

EDUCATION

B.S. - ROBOTICS ENGINEERING

WORCESTER POLYTECHNIC INSTITUTE

Expected Graduation May 2027

B.S. - ELECTRICAL AND COMPUTER ENGINEERING

WORCESTER POLYTECHNIC INSTITUTE

Expected Graduation May 2027